

Section A: Pure Mathematics [35 marks]

- 1** (i) Find $\int \frac{1}{(2x+3)^5} + \frac{2}{2x+3} \, dx$. [2]
- (ii) Find $\int \sqrt{5x-2} + e^{2x} \, dx$. [2]
- (iii) Evaluate $\int_1^{-1} e^{x^2} \, dx$. [1]
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- 2** (i) Differentiate $(2x^2 + 3)^{\frac{3}{2}} - \ln(3x-1)$ with respect to x . [3]
- (ii) Hence find the area bounded by the curves $y = x(2x^2 + 3)^{\frac{1}{2}}$, $y = \frac{1}{6x-2}$, and the lines $x=1$ and $x=2$ in the form $p\sqrt{11} + q\sqrt{5} + r \ln\left(\frac{2}{5}\right)$, where p , q , and r are constants to be determined. [3]
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- 3** (i) Let $f(x) = x^4 + x^3 - x^2 - x - 2$. Sketch the graph of $y = f(x)$, showing clearly all stationary points and axial intercepts. [3]
- (ii) Hence state the range of values of x for which $f'(x) \leq 0$. [1]

The equations of C_1 and C_2 are given as follows:

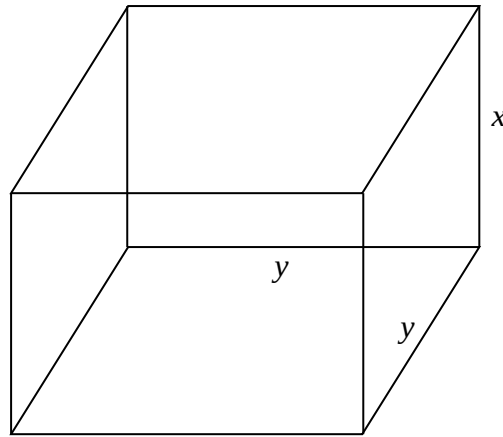
$$C_1 : y = x^4 + 2x^3 - x - 1$$

$$C_2 : y = x^3 + x^2 + 1$$

- (iii) Show that the intersection(s) of the graphs C_1 and C_2 satisfy the equation $x^4 + x^3 - x^2 - x - 2 = 0$. [1]
- (iv) Hence find the coordinates of the intersection point(s) of C_1 and C_2 . [2]

- 4 (a) Find the value(s) of k for which $kx^2 + kx - 1 = 0$, $k \neq 0$, has real and equal roots, and solve for x , for these value(s) of k . [4]

(b)



The dimensions of a box open at the top are as shown above. It is known that the total external surface area of such a box is twice its base area. Formulate an equation relating x and y . [1]

Given also that $2x + y = 1$, solve for x and y in exact form. [3]

- 5 (a) The volume of a square pyramid with base length b cm and height a cm is 10 cm^3 .
Find a in terms of b . [Volume of pyramid is given by $\frac{1}{3} \times \text{base area} \times \text{height}$] [1]
When $a = 1$, a is decreasing at a rate of 0.05 cm s^{-1} . Calculate the rate of change of b at the same point in time. [4]
- (b) The normal to the curve $y = e^x + e^{-x}$ at $x = \ln 2$ cuts the curve again at $x = t$. Find the value of t , correct to three decimal places. [4]

Section B: Statistics [60 marks]

- 6 For events A and B , it is given that $P(A) = \frac{2}{5}$, $P(B) = \frac{7}{10}$ and $P(A' | B) = \frac{23}{35}$.
Find
- (i) $P(A' \cap B)$, [2]
- (ii) $P(A \cup B)$. [2]
- A third event C is given such that $P(C) = \frac{5}{7}$, and that B and C are independent.
- (iii) Find $P(B \cap C)$ [2]
- (iv) Hence state an inequality satisfied by $P(A \cap B \cap C)$. [2]

- 7** Nelly is a shot-put thrower in her school and has been training for a long time. The distance, x metres, that she can throw her shot-put is found to have a mean of 19.6 metres and a standard deviation of 0.51 metres. At the beginning of the current school year, a new coach took over and started using a new training method on her which is supposed to boost her performance.

After several weeks of training, the distances she threw on 55 randomly selected attempts were recorded as $\sum x = 1085.15$. A test is to be carried out to determine whether her performance has improved under the new training method.

- (i) State clearly the null hypothesis and the alternative hypothesis that you are using, defining any symbols used. [2]
- (ii) Test, at the 5% significance level, whether the new training method has improved her performance. [4]
- (iii) Explain whether it is necessary to assume that X has a normal distribution in order for the test in part (ii) to be valid. [1]
- (iv) Explain, in context of the question, the meaning of “at the 5% significance level”. [1]

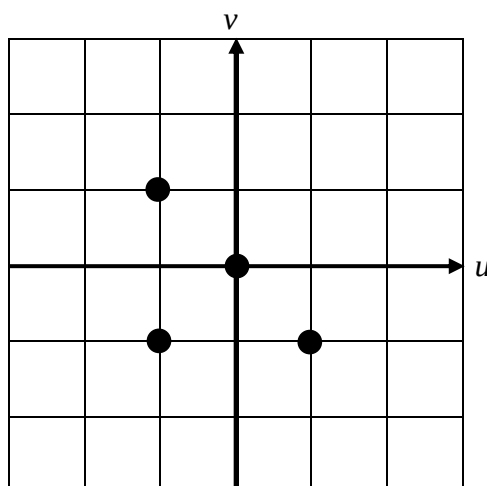
- 8** Over a long period of time, it is found that 6% of camera lenses manufactured by Kadok Company are faulty.

- (i) Find the probability that, out of 100 camera lenses manufactured by Kadok, at most 7 are faulty. [2]
- (ii) Find the probability that, out of 100 camera lenses manufactured by Kadok, more than 2 are faulty, given that there are at least 93 non-faulty camera lenses. [3]

Camera lenses are packed in boxes of 100 for shipping to companies that use them to make their cameras.

- (iii) Using a suitable approximation, find the probability that, out of 100 boxes of camera lenses ready for shipping, more than 40, but at most 65 have less than 8 faulty lenses. [4]

- 9 (a) The diagram below shows four pairs of bivariate data (u , v) plotted on a scatter plot, denoted by dots, giving a negative linear product moment correlation coefficient.



A fifth pair of data is to be placed on the scatter plot. For each of the cases below, copy the diagram above and indicate the fifth pair of data on the diagram using a '×', showing that the five pairs of data resulting in a linear product moment correlation coefficient that is

- (i) positive,
(ii) zero. [2]
- (b) Kelvin decided to carry out an investigation to study the relationship between the time taken by a male student to complete the 2.4 km run, x mins, and the distance the male student can jump in the Standing Broad Jump, y m, of the Physical Fitness Test conducted yearly in local junior colleges. Data from seven individuals were collected and shown in the table below.

x (mins)	14.2	13.9	12.9	11.6	11.2	9.45	9.05
y (m)	1.90	1.95	1.99	2.19	2.24	2.36	2.43

- (i) Determine the value of the product moment correlation coefficient between y and x . Comment on the value of the product moment correlation coefficient in the context of the question. [2]
- (ii) Using a suitable regression line, obtain an estimate of the time taken to complete the 2.4 km run if a male student jumped a distance of 2.28 m for the standing broad jump. [2]
- (iii) A set of (a, b) data was obtained from an eighth male student, and was added to the seven pairs of data above. The new regression line of y on x is given as $y = -0.08581x + 3.12473$, and the new regression line of x on y is given as $x = -8.25664y + 29.16979$. Determine the values of the new means of x and y and hence find a and b . [4]

10 An IP school has 2000 students – there are 200 students in each level from Secondary 1-4, 600 in JC1 and 600 in JC2. The Student Council is interested in finding $\bar{x}$ hours, the length of time each student spends in a week doing homework. The president of the Student Council thus decides to survey 50 students.

- (i) Describe how she can do this using
- (a) stratified sampling, [2]
 - (b) quota sampling. [2]
- (ii) In the context of the question, state an advantage of choosing
- (a) stratified sampling over quota sampling, [1]
 - (b) quota sampling over stratified sampling. [1]

The data gathered is summarised as follows:

$$\sum x = 600; \quad \sum x^2 = 7300$$

- (iii) (a) Calculate the unbiased estimates of the population mean and variance. [2]
- (b) State whether your calculated estimates will differ from the sample mean and sample variance. If it does, determine the differing sample value. [2]
- (iv) Assuming that the population mean is the value of the unbiased estimate found in (iii)(a), find the probability that the sample mean exceeds 12.1. [2]

- 11** [In this question, state clearly the appropriate parameters of any distribution you use in your calculations.]

Valet parking is a service by which cars are parked by service staff. At a shopping centre which offers an optional valet parking service for a fee, it is found that on a weekday, the length of time a customer leaves his vehicle in the car park has a normal distribution with mean 95 minutes and standard deviation 21 minutes.

- (i) Find the probability that a randomly selected customer leaves his vehicle in the car park on a weekday for more than 120 minutes. [1]
- (ii) Find the probability that on a weekday, out of two randomly selected customers who leave their separate vehicles in the car park, one leaves it for more than 120 minutes and the other leaves it for between 80 to 112 minutes. [2]

The weekday parking charges per vehicle is \$0.04 per minute, plus an additional \$2 each time you use their valet parking service. Mr. Ling drove to the shopping centre once a day on two weekdays, and used the valet parking service on both occasions.

- (iii) Find the probability that Mr. Ling paid a total of less than \$9 on the two weekdays. [3]

50 customers left their personal vehicles in the car park of the shopping centre during one weekday.

- (iv) Find the probability that there are exactly 10 customers who leave their vehicles in the car park for more than 120 minutes. [2]

On a weekend, the length of time a customer leaves his vehicle in the car park has an independent normal distribution with mean μ minutes and standard deviation σ minutes. It is given that the probability of a customer leaving his vehicle in the car park on a weekend for at most 108 minutes and at least 148 minutes are 0.354 and 0.191 respectively.

- (v) Find the values of μ and σ , both corrected to the nearest minute. [5]

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